



CogView3文到图生成技术介绍

2024年2月 智谱AI技术团队

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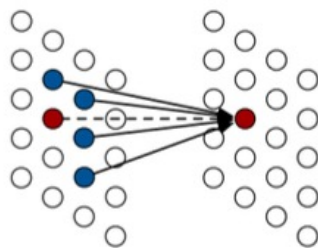
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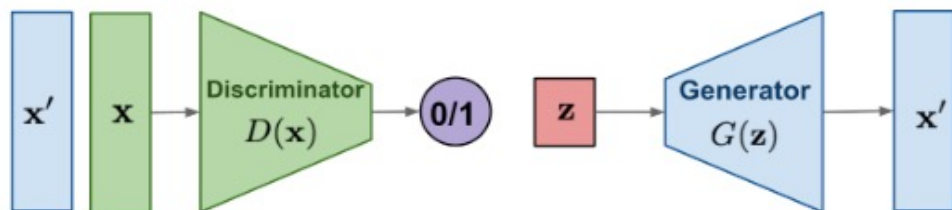
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生成模型比较

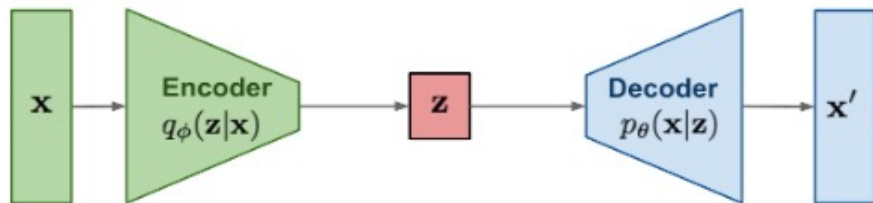
AR: Autoregressive models:



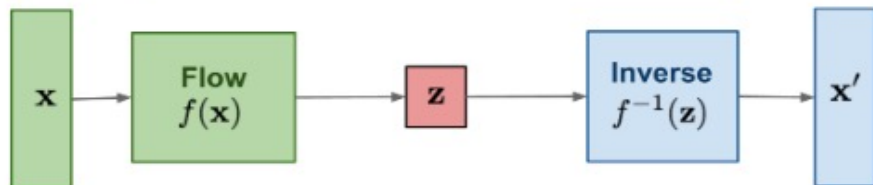
GAN: Adversarial training



VAE: maximize variational lower bound



Flow-based models:
Invertible transform of distributions



Diffusion models:
Gradually add Gaussian noise and then reverse



过去的生成模型存在的一些局限性:

- 推理时间过长 (**AR**)
- 分布受限且训练困难 (**GAN**)
- 需要存在可逆变换 (**FM**)

与之前的生成模型相比,

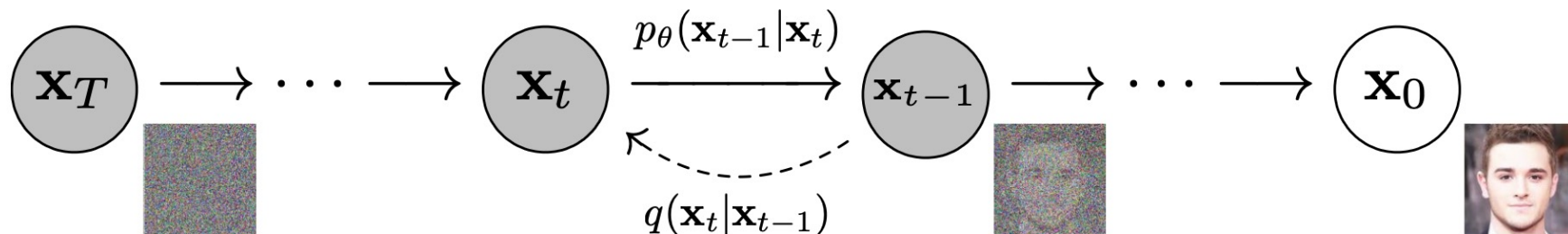
Diffusion Model具有:

- **更强的广域分布建模能力**
- **较好的细节生成能力**
- **更易训练**

Diffusion Model原理: DDPM

DDPM: Denoising Diffusion Probabilistic Models

逐步添加高斯噪声 (图像建模) ↔ 逐步去除高斯噪声 (图像生成)



前向过程 (逐步添加噪声) :

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}), \quad q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

是马尔可夫过程, 其中 $\mathbf{x}_0$ 代表真实图像, $1e^{-4} = \beta_1 < \dots < \beta_T = 0.02$ 线性增长

令 $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, 则有:

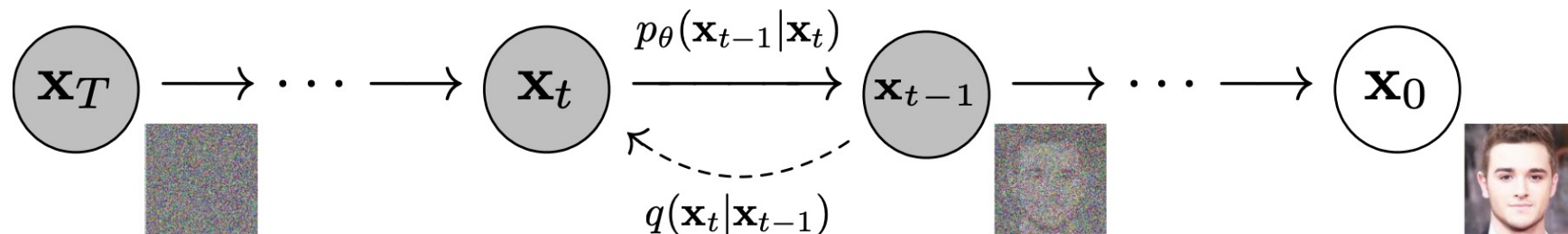
$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1 - \alpha_t}\epsilon_{t-1} = \dots = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

因此, 训练时中间状态 $\mathbf{x}_t$ 可直接采样随机高斯噪声获得

Diffusion Model原理: DDPM

DDPM: Denoising Diffusion Probabilistic Models

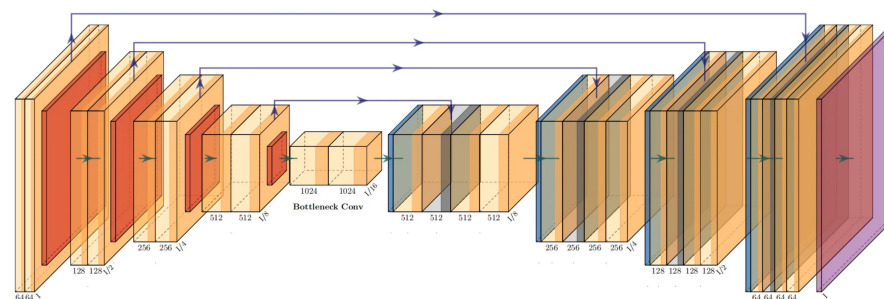
逐步添加高斯噪声 (图像建模) ↔ 逐步去除高斯噪声 (图像生成)



反向过程 (逐步去除噪声) : 通过神经网络 p_θ 拟合后向分布 $p(\mathbf{x}_{t-1}|\mathbf{x}_t)$

当 β 足够小时, 后向分布也可以近似视为 Gaussian 分布:

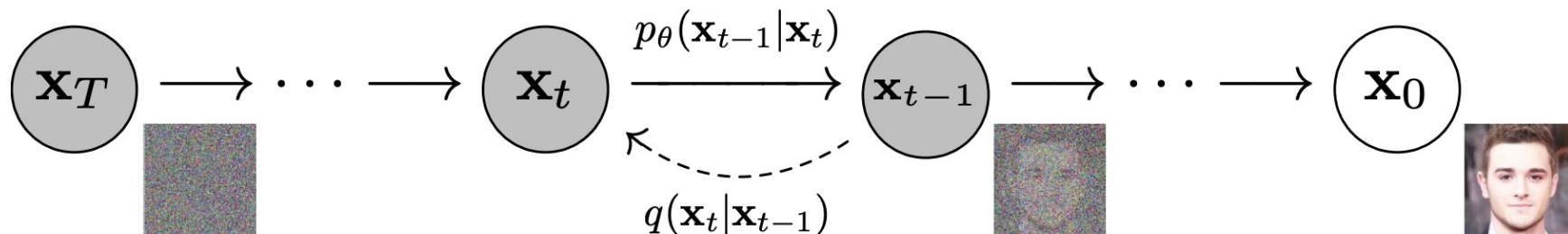
$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)}_{\text{神经网络建模 (UNet)}}, p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$$



Diffusion Model原理：DDPM

DDPM: Denoising Diffusion Probabilistic Models

逐步添加高斯噪声（图像建模） ↔ 逐步去除高斯噪声（图像生成）



损失函数构建：基于KL散度

当 $\mathbf{x}_0$ 已知时，易给出条件后向分布：

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I})$$

其中, $\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_{t-1}}\beta_t}{1 - \bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t$, $\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}\beta_t$

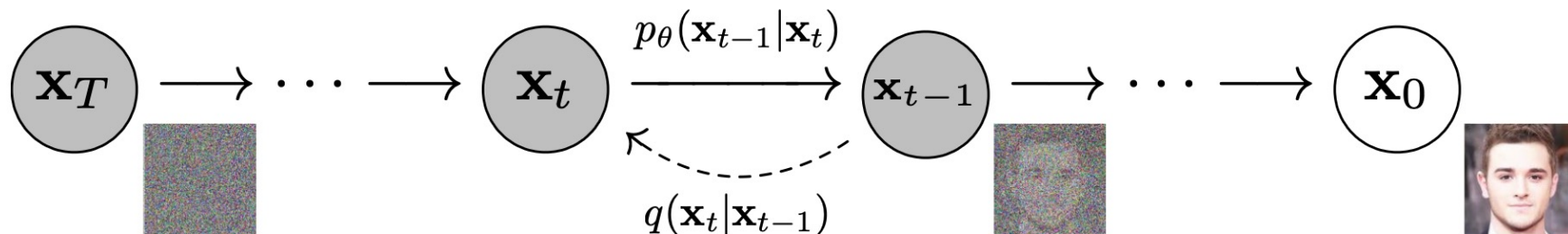
基于KL散度，进行预测分布对真实分布的拟合：

$$\mathbb{E}_q[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T))}_{L_T} + \sum_{t=2}^T \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \underbrace{\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}]$$

Diffusion Model原理: DDPM

DDPM: Denoising Diffusion Probabilistic Models

逐步添加高斯噪声 (图像建模) ↔ 逐步去除高斯噪声 (图像生成)



损失函数构建: 基于KL散度

$$\mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T))}_{L_T} + \sum_{t=2}^T \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right]_{L_0}$$

固定 $\Sigma_\theta = \sigma_t^2 \mathbf{I}$ 为常数, 则 $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma_t^2 \mathbf{I})$, 代入得

$$L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2 \right] + C$$

使用 ϵ -prediction 得

$$\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}_t(\mathbf{x}_t, \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t))) = \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t))$$

Diffusion Model原理: DDPM

DDPM: Denoising Diffusion Probabilistic Models is a learned gradient estimator, with the parameterization (11), Eq. (10) simplifies to:

逐:

$$\mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2 \right]$$

$\mathbf{x}_T$

→ Denoising score matching over multiple noise scales indexed by t [55]. A (in the term of) the variational bound for the Langevin-like reverse process (1) (Using an objective resembling denoising score matching is equivalent to using the finite-time marginal of a sampling chain resembling Langevin dynamics).

损失函数构建: 基于KL散度

代入 ϵ -prediction的两个等式, 得到最终的loss函数如:

$$\mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2 \right]$$

在实践中, 作者发现权重项简化的loss函数效果更好, 因此使用了权重简化的loss形式:

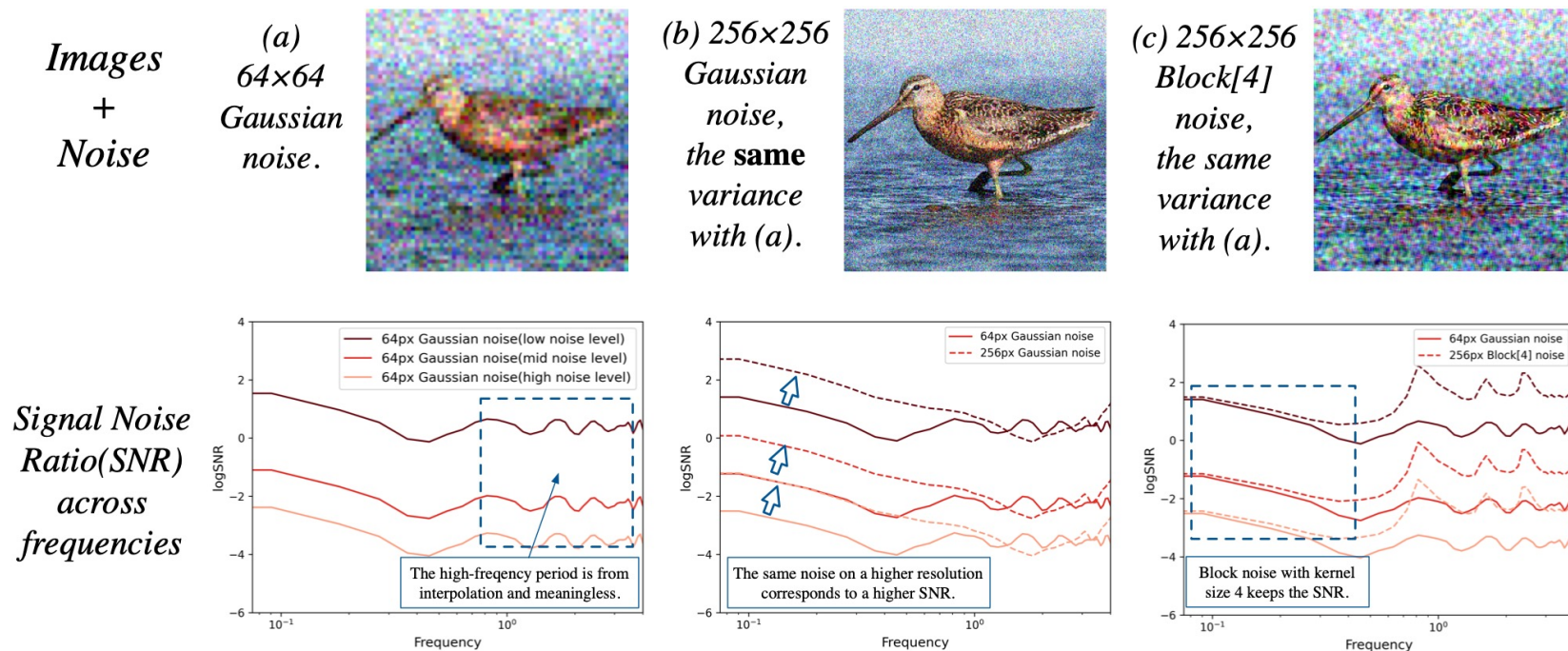
$$\mathbb{E}_{\mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2]$$

Diffusion Model原理: Relay Diffusion

Relay Diffusion: Unifying diffusion process across resolutions for image synthesis

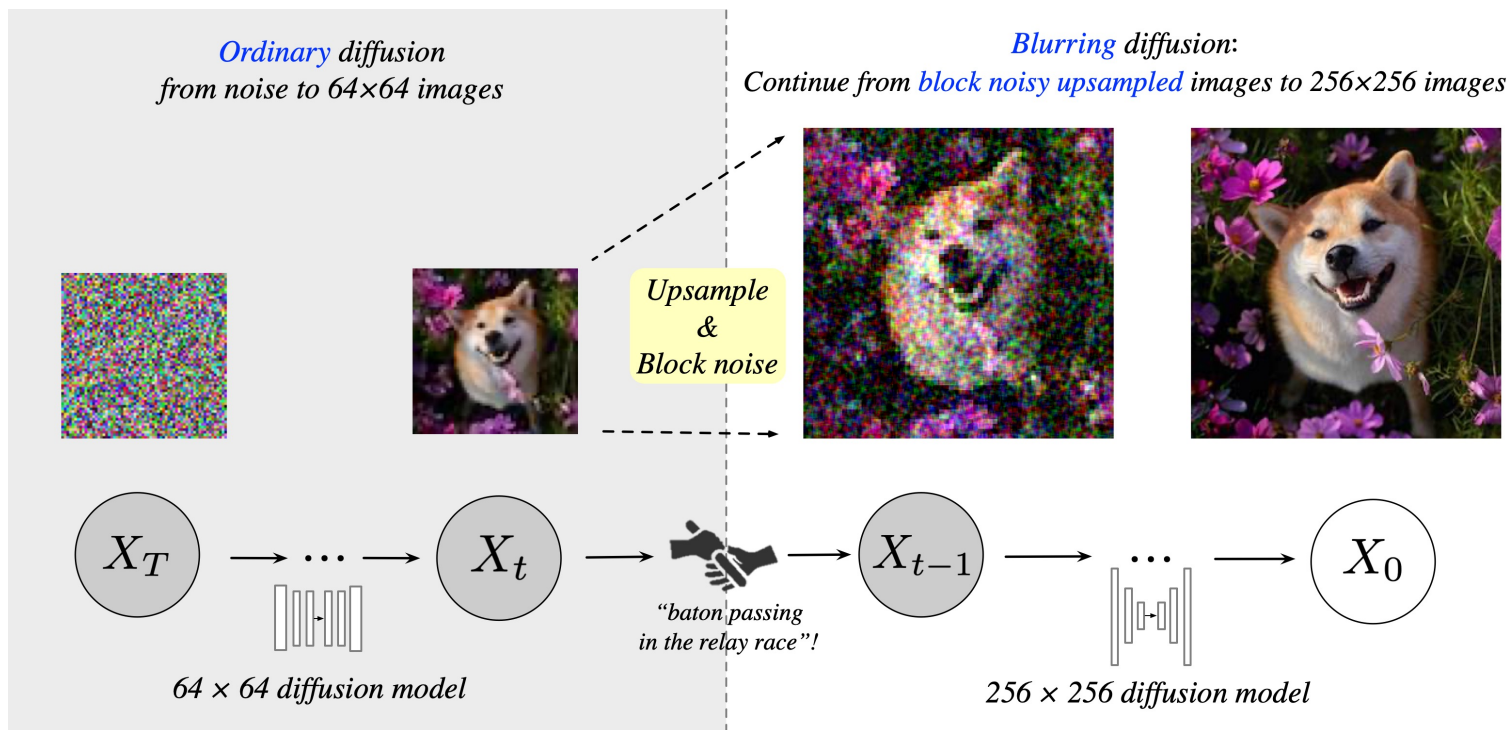
目前的Diffusion Model框架依然存在一些问题

- 训练及推理的成本较高（收敛较慢&步数较多）
- 目前使用的Noise schedule（即逐步添加噪声的比例分配）可能并不合理
- 高斯噪声不利于建模图像频域信息等特征



Diffusion Model原理: Relay Diffusion

Relay Diffusion: Unifying diffusion process across resolutions for image synthesis



$$Block[s](\epsilon)_{x,y} = \frac{1}{s} \sum_{i=0}^{s-1} \sum_{j=0}^{s-1} \epsilon_{x-i,y-j}$$

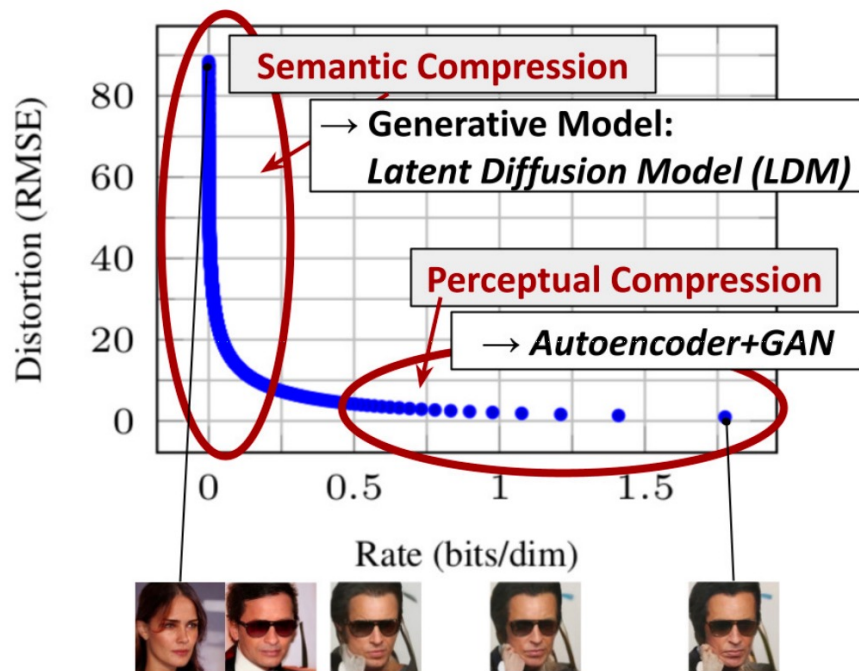
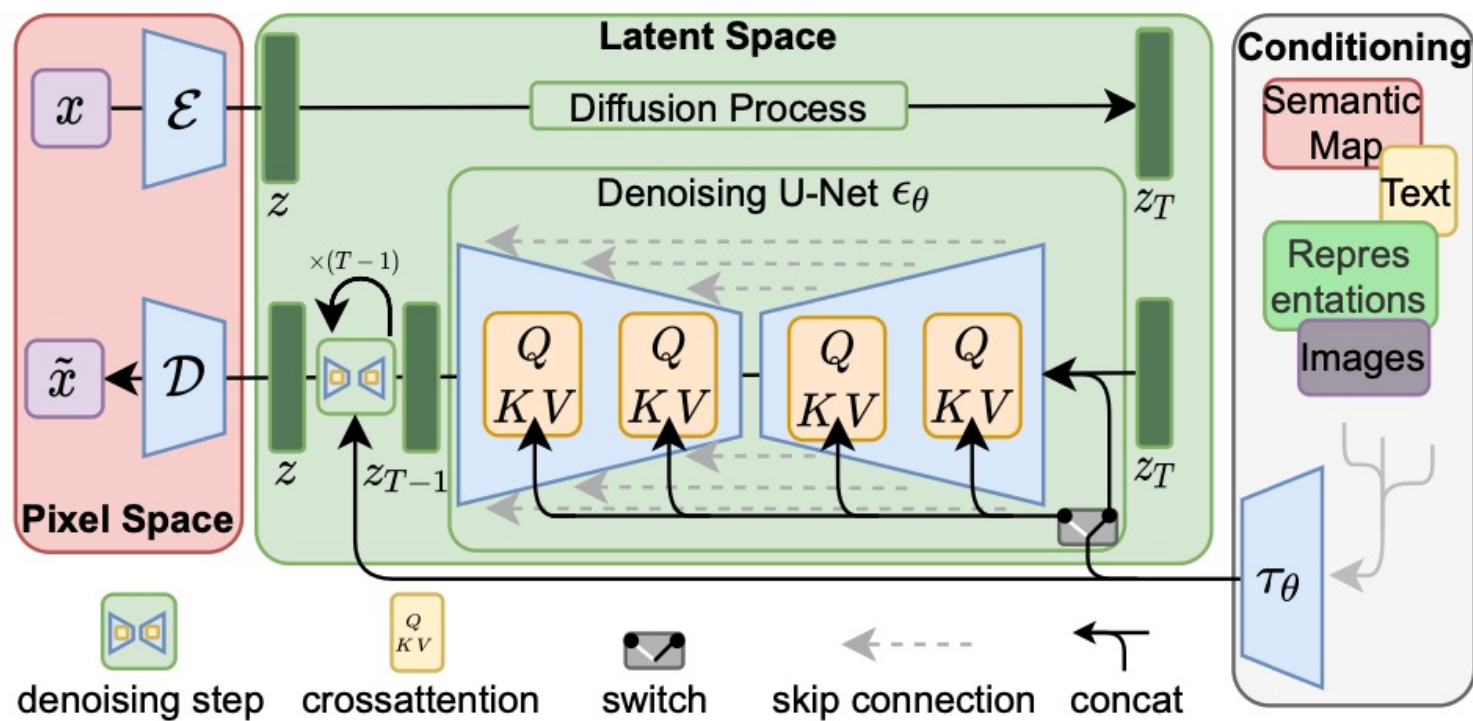
$$dis(x_0, x_1) = \min(|x_0 - x_1|, x_{max} - |x_0 - x_1|)$$

根据存在的问题给出的解决方案

- 分阶段建模生成：低分辨率阶段降低成本；超分辨率接力（Relay）生成，具备修正不佳细节的能力
- 引入不同位置间存在相关性的Block Noise，更好的衔接并建模频域信息

从Latent Diffusion到SDXL

High-Resolution Image Synthesis with Latent Diffusion Models



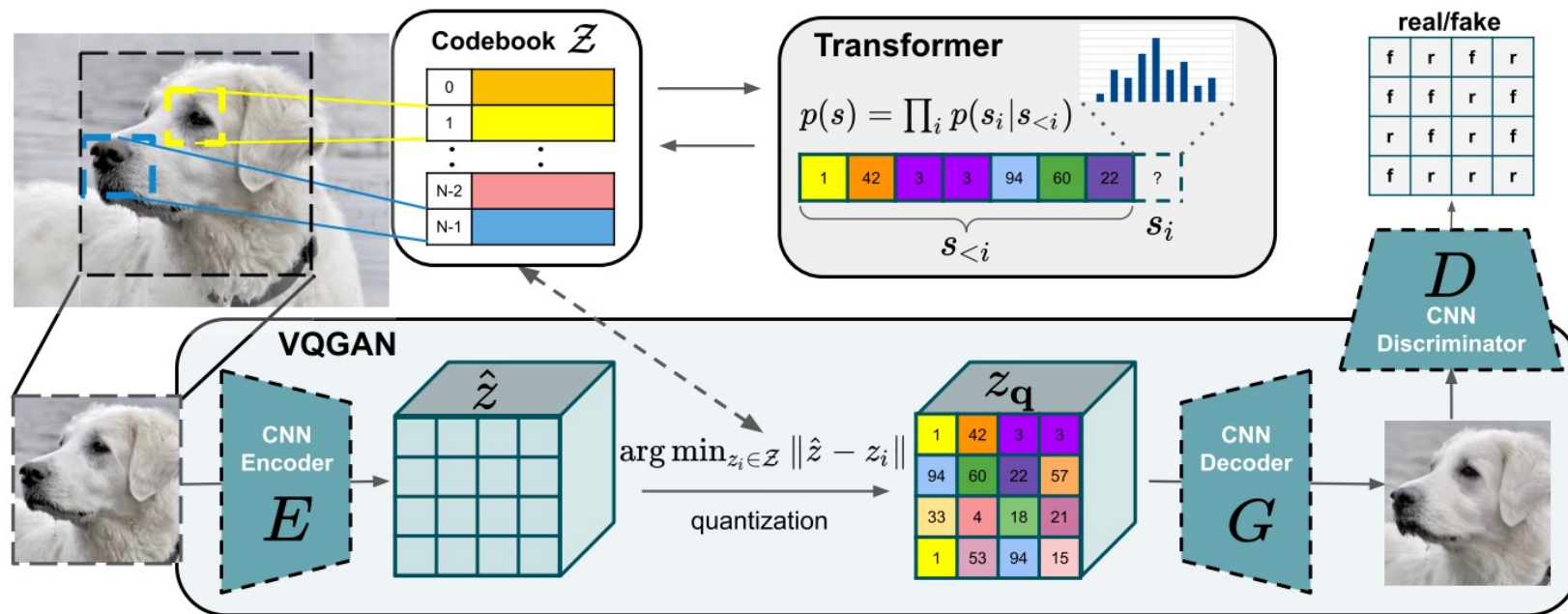
使用自编码器将图片从像素空间压缩至隐空间:

$$z = \varepsilon(x), \tilde{x} = \mathcal{D}(z), \quad x \in \mathbb{R}^{H \times W \times C}, z \in \mathbb{R}^{h \times w \times c}$$

其中图像分辨率压缩 $f = H/h = W/w = 8$ 倍, 极大的降低了训练的成本及难度

从Latent Diffusion到SDXL

High-Resolution Image Synthesis with Latent Diffusion Models



AutoEncoder的选择：VQGAN或KL regularization

训练损失函数：

$$L_{AE} = \min_{\varepsilon, \mathcal{D}} \max_{\psi} \underbrace{(L_{rec}(x, \mathcal{D}(\varepsilon(x))) - L_{adv}(\mathcal{D}(\varepsilon(x))) + \log \mathcal{D}_{\psi}(x))}_{\text{GAN Loss}} + \underbrace{L_{reg}(x; \varepsilon, \mathcal{D})}_{\text{隐空间Reg Loss (VQ / KL)}}$$

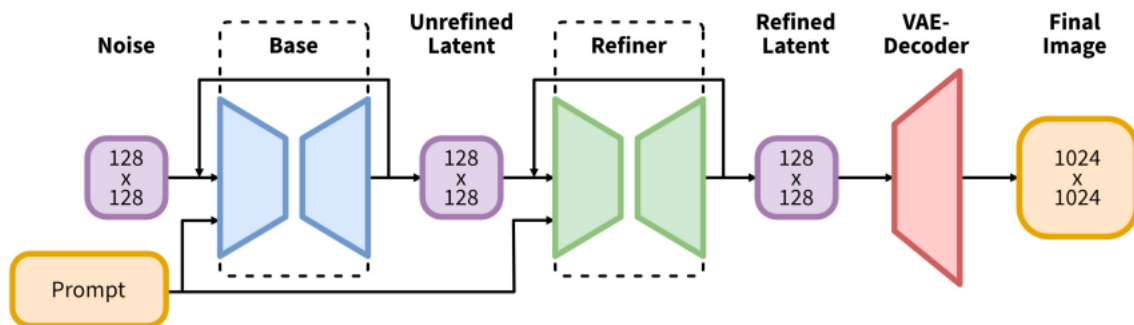
重建Loss
(L1 / L2)

GAN Loss

隐空间Reg Loss
(VQ / KL)

从Latent Diffusion到SDXL

SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis



添加Crop Conditioning,
缓解了生成图片裁剪问题:



文本到图像生成的Diffusion Model

- 使用Text encoder (CLIP text encoder) 获得文本对应的embedding
- 添加Cross Attention模块输入文本信息
- Classifier-free Guidance (CFG) 增强语义控制

SDXL训练了参数量2.9B的Text-to-Image Latent Diffusion模型, 得到了开源模型中最佳的生成效果。

Multi-aspect Training
以支持不同形状生成:

Height	Width	Aspect Ratio	Height	Width	Aspect Ratio
512	2048	0.25	1024	1024	1.0
512	1984	0.26	1024	960	1.07
512	1920	0.27	1088	960	1.13
512	1856	0.28	1088	896	1.21
576	1792	0.32	1152	896	1.29
576	1728	0.33	1152	832	1.38
576	1664	0.35	1216	832	1.46
640	1600	0.4	1280	768	1.67
640	1536	0.42	1344	768	1.75
704	1472	0.48	1408	704	2.0
704	1408	0.5	1472	704	2.09
704	1344	0.52	1536	640	2.4
768	1344	0.57	1600	640	2.5
768	1280	0.6	1664	576	2.89
832	1216	0.68	1728	576	3.0
832	1152	0.72	1792	576	3.11
896	1152	0.78	1856	512	3.62
896	1088	0.82	1920	512	3.75
960	1088	0.88	1984	512	3.88
960	1024	0.94	2048	512	4.0

DALL-E-3: 图像重概括 (recaption)

Improving Image Generation with Better Captions

在模型结构、训练的探索之外，数据的质量对文本到图像生成模型的训练也至关重要

DALL-E-3基于GPT-4V对于图像进行描述，作为训练数据95%的caption:

Image			
Alt Text	now at victorian plumbing.co.uk	is he finished...just about!	23 (19 of 30) 1200
SSC	a white modern bathtub sits on a wooden floor.	a quilt with an iron on it.	a jar of rhubarb liqueur sitting on a pebble background.
DSC	this luxurious bathroom features a modern freestanding bathtub in a crisp white finish. the tub sits against a wooden accent wall with glass-like panels, creating a serene and relaxing ambiance. three pendant light fixtures hang above the tub, adding a touch of sophistication. a large window with a wooden panel provides natural light, while a potted plant adds a touch of greenery. the freestanding bathtub stands out as a statement piece in this contemporary bathroom.	a quilt is laid out on a ironing board with an iron resting on top. the quilt has a patchwork design with pastel-colored strips of fabric and floral patterns. the iron is turned on and the tip is resting on top of one of the strips. the quilt appears to be in the process of being pressed, as the steam from the iron is visible on the surface. the quilt has a vintage feel and the colors are yellow, blue, and white, giving it an antique look.	rhubarb pieces in a glass jar, waiting to be pickled. the colors of the rhubarb range from bright red to pale green, creating a beautiful contrast. the jar is sitting on a gravel background, giving a rustic feel to the image.

DALL-E-3：图像重概括（recaption）

Improving Image Generation with Better Captions

生成图像时，使用GPT-4对用户prompt进行改写（prompt upsample），获得描述详细且与训练数据更对齐的画图指令：

A bird scaring a scarecrow.

A large, vibrant bird with an impressive wingspan swoops down from the sky, letting out a piercing call as it approaches a weathered scarecrow in a sunlit field. The scarecrow, dressed in tattered clothing and a straw hat, appears to tremble, almost as if it's coming to life in fear of the approaching bird.

Paying for a quarter-sized pizza with a pizza-sized quarter.



A person is standing at a pizza counter, holding a gigantic quarter the size of a pizza. The cashier, wide-eyed with astonishment, hands over a tiny, quarter-sized pizza in return. The background features various pizza toppings and other customers, all of them equally amazed by the unusual transaction.

A smafml vessef epropoeilled on watvewr by ors, sauls, or han engie.

A small vessel, propelled on water by oars, sails, or an engine, floats gracefully on a serene lake. the sun casts a warm glow on the water, reflecting the vibrant colors of the sky as birds fly overhead.

基于训练数据的优化，
DALL-E-3模型的语义相关性
得到了显著提升，
人类偏好程度也更高：

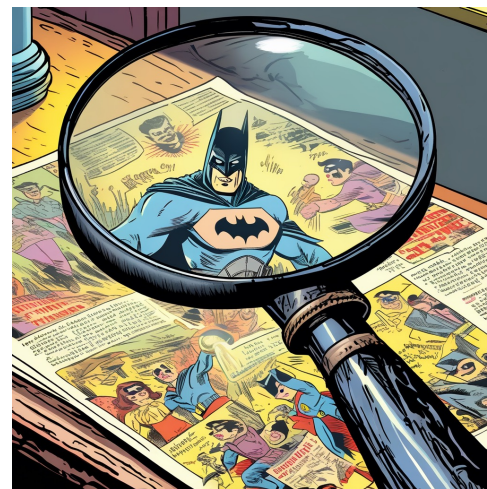
Dataset	DALL-E 3	Midjourney 5.2	Stable Diffusion XL	DALL-E 2
DALL-E 3 Eval (prompt following)	153.3	-104.8	-189.5	-
DALL-E 3 Eval (style)	74.0	30.9	-95.7	-
MSCOCO (coherence)	71.0	48.9	-84.2	-
Drawbench	61.7	-	-34.0	-79.3



一只乌龟坐在森林里的
鱼眼镜头



花椰菜下的斑马



一页50年代的蝙蝠侠漫画上
的一个放大镜

一只羊驼戴着墨镜站在宇宙
飞船的甲板上，背景是地球



长着叶子的手掌



时代广场滑板上的泰迪熊



CogView3: 样例展示

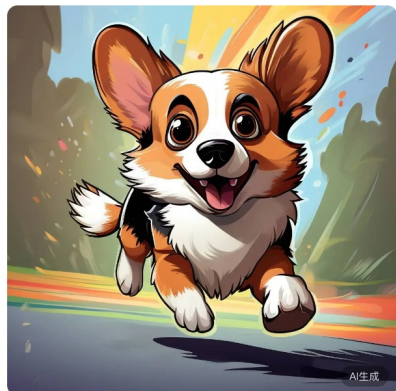
画一个卡通柯基

AI绘画：已完成



把这个柯基换成在跑步

AI绘画：已完成



跑得再快一些

AI绘画：已完成



再快一些!

AI绘画：已完成



更快一些!

AI绘画：已完成



CogView3: 使用演示

通过智谱清言官网或APP (GLM-4 all-tools) 使用CogView3
访问<https://chatglm.cn/main/alltoolsdetail>



CogView3: 使用演示

通过GLM-4 API接口使用CogView3

参考<https://open.bigmodel.cn/dev/api#cogview>

CogView

模型编码: cogview-3

根据用户的文字描述生成图像,使用同步调用方式请求接口

创建文生图模型同步请求

接口请求

传输方式	HTTPS
请求地址	https://open.bigmodel.cn/api/paas/v4/images/generations
调用方式	同步调用, 等待模型执行完成并返回最终结果
字符编码	UTF-8
接口请求格式	JSON
响应格式	JSON
接口请求类型	POST
开发语言	任意可发起 HTTP 请求的开发语言

接口请求参数

参数名称	类型	是否必填	参数说明
model	String	是	模型名称
prompt	String	是	所需图像的文本描述

接口响应参数

参数名称	类型	参数说明
created	String	请求创建时间, 是以秒为单位的Unix时间戳。
> data	List	数组, 包含生成的图片 URL。目前数组中只包含一张图片。

Python 调用示例

请求示例

```
1 from zhipuai import ZhipuAI
2 client = ZhipuAI(api_key="") # 请填写您自己的APIKey
3
4 response = client.images.generations(
5     model="cogview-3", #填写需要调用的模型名称
6     prompt="一只可爱的小猫咪",
7 )
8 print(response.data[0].url)
```

响应示例

```
1 {
2   "created": 1703485556,
3   "data": [
4     {
5       "url": "https://....."
6     }
7   ]
8 }
```



感谢您的聆听

2024年2月 智谱AI技术团队

北京智谱华章科技有限公司